Chapter 33 — Conflict Resolution & a Sync Engine

> **Part IV — Offline-First & Synchronization** · Complexity: **Complex** (≥40 interview questions)
>
> **Prerequisites:** Ch 29 (offline-first), Ch 31 (outbox/push), Ch 32 (delta/pull), Ch 30 (WorkManager), Ch 15 (idempotency), Ch 3 (SSOT). This chapter **completes Part IV** by resolving the divergence that push+pull create, and composes the pieces into one **sync engine**.
>
> **Reference surface in StreamCart:** reconciling edits made on two devices (or offline + server) to the same cart item, wishlist, watch-progress, or profile — deterministically and without losing user intent.

33.1 Problem Definition and Real-World Context

Once writes happen **both** locally (offline, Ch 31) and remotely (other devices/ users, Ch 32), the same entity can be changed in two places before they sync — a **conflict**. Without a resolution policy, sync either **loses data** (one change silently overwrites the other) or **corrupts state** (partial/inconsistent merges). Conflict resolution is the hardest part of sync and the thing interviewers probe:

- **Silent last-writer-wins by accident.** A naive "upsert whatever arrived last" overwrites a concurrent edit — the user's change vanishes with no indication.
- **Clock-based LWW is subtly wrong.** Using wall-clock timestamps to pick a winner fails under clock skew and can't express causality ("did B see A's change?").
- **Field-level loss.** Entity-level overwrite discards *all* fields when only one field conflicts (two users edit different fields → one edit lost unnecessarily).
- **No convergence guarantee.** Devices that apply conflicting changes in different orders can end up in **different** final states (not eventually consistent).
- **Ignoring user intent.** For important edits (a document, a shared list), silently auto-resolving may be wrong; the user may need to choose.

The architecture: **detect** conflicts (versions / vector clocks), **resolve** by a policy appropriate to the data (last-writer-wins, server-wins, field-merge, operation-merge/**CRDT**, or **user-resolved**), guarantee **convergence** (all devices reach the same state), and compose detection/resolution with push (Ch 31) + pull (Ch 32) into a coherent **sync engine**. This chapter is the capstone that makes offline-first *correct*, not just *available*.

Three practical use cases

1. **E-commerce (StreamCart).** Cart quantity edited on phone (offline) and web (online); resolve by LWW or sum-merge depending on semantics — don't lose the user's intent.
2. **Productivity/Social (Social).** A shared list or draft edited on two devices; **field-level merge** or **CRDT** so both edits survive; user-resolved for true conflicts.
3. **OTT (Watch).** Watch-progress on two devices; **max-progress** merge (a simple CRDT) so the furthest position wins deterministically.

Why this is architectural

Conflict resolution defines the app's **consistency guarantees** (convergence, eventual consistency), the **data-loss policy** (what happens when edits collide), and the **merge model** (entity/field/operation/CRDT/user) — the properties that decide whether multi-device offline-first data stays correct.

33.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Detect when local and remote versions of an entity diverge.
F2	Resolve conflicts by a per-data-type policy (LWW / server-wins / merge / user).
F3	Guarantee convergence: all replicas reach the same state regardless of order.
F4	Preserve causality where it matters (versions/vector clocks, not wall clocks).
F5	Support field-level and operation-level merge, not just entity overwrite.
F6	Surface true conflicts to the user when auto-resolution is unsafe.
F7	Compose with push (Ch 31) + pull (Ch 32) into one sync engine.

Non-functional requirements

Attribute	Policy
Consistency	Convergent (eventually consistent); deterministic resolution.
Correctness	No silent data loss where avoidable; causality respected.
UX	Auto-resolve safely; prompt only for genuine conflicts.
Performance	Resolution cheap; merge scalable.
Observability	Conflict rate + resolution outcomes visible.

Scope

- **In scope:** conflict detection (version/vector clocks), resolution strategies (LWW, server/client-wins, field-merge, CRDT/op-merge, user-resolved), convergence, three-way merge, the composed sync engine (detect → resolve → apply), conflict UX.
- **Out of scope:** push/outbox mechanics (Ch 31), delta/token mechanics (Ch 32), WorkManager (Ch 30), real-time transport (Ch 63).

Assumptions & constraints

- Entities carry a **version** (and optionally a vector clock / `updatedAt` as a tiebreaker). Backend participates (server version, merge endpoints where needed). Room SSOT (Ch 3). `minSdk 24`. Eventual consistency

acceptable.

33.3 System Design Fundamentals

- **Conflict detection.** A conflict exists when **both** the local replica and the remote changed the same entity since they last agreed. Detect via **versions**: the client holds `baseVersion` (what it last synced); if the server's version advanced *and* the client has a pending local change, they diverged. **Vector clocks** (a per-replica counter map) capture *causality* precisely — whether one change "happened-before" another or they're truly concurrent — where plain versions can't.
- **Why not wall-clock LWW?** Timestamps can't express causality and suffer clock skew; two devices with wrong clocks pick the "wrong" winner. Use **logical versions/vector clocks**; if you use timestamps, use them only as a deterministic **tiebreaker** on the server's clock.
- **Resolution strategies (choose per data type):**
  - **Last-Writer-Wins (LWW):** pick the higher version/timestamp. Simple, convergent, but **loses** the other edit. OK for single-value, low-stakes fields.
  - **Server-wins / client-wins:** always prefer one side. Simple; biased data loss.
  - **Field-level (three-way) merge:** compare `base`, `local`, `remote` per field; take non-conflicting field changes from both; only truly-conflicting *same* fields need a tiebreak. Preserves more intent.
  - **Operation-based merge / CRDTs:** model changes as commutative operations so any order converges without conflict — e.g., **G-Counter** (sum), **LWW- Register**, **OR-Set** (add/remove set), **max** (watch-progress). CRDTs guarantee convergence *by construction*; best for collaborative/mergeable data.
  - **User-resolved:** present both versions and let the user choose/merge. For high-value edits where auto-resolution is unacceptable.
- **Convergence (the correctness bar).** All replicas must reach the **same** final state regardless of the order changes arrive. LWW and CRDTs are convergent; ad-hoc merges often aren't. This is what "eventual consistency" actually requires.
- **Three-way merge** (base/local/remote) is the key to not losing concurrent edits to different fields — like a VCS merge.
- **The sync engine** composes: **push** local changes (outbox, Ch 31) → **pull** remote deltas (Ch 32) → on each incoming change, **detect** conflict vs local pending → **resolve** by policy → apply to SSOT (Ch 3). Idempotency (Ch 15) and versions keep it safe under retries.

Core insight: **detect conflicts with logical versions/vector clocks, resolve with a strategy matched to the data's semantics (LWW for single values, field-merge or CRDTs for collaborative data, user-resolved for high-stakes), and guarantee all replicas converge** — then compose detect→resolve with push/pull into one engine.

33.4 High-Level Architecture


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Components & ownership

- **Sync engine** orchestrates push→pull→detect→resolve→apply.
- **Detector** compares versions/vector clocks (local pending vs remote).

- **Resolver** applies the per-type policy (LWW / merge / CRDT / user).
- **SSOT (Room)** stores the resolved entity + version + status (Ch 3/29).
- **UI** handles user-resolved conflicts.

Direction: pull delta → detect vs local pending → resolve (auto or user) → apply to SSOT (convergent).

33.5 Detailed Component Design



Component	Strategy	Convergent?	Data loss	Use for
`LwwResolver`	highest version/ts wins	yes	yes (other edit)	single low-stakes values
server/client-wins	fixed side	yes	yes (biased)	authoritative/local-trusted fields
`FieldMergeResolver`	three-way per field	yes (deterministic)	minimal	structured records
`CrdtResolver`	commutative ops	yes (by construction)	none	collaborative/mergeable (sets, counters, progress)
`UserPromptResolver`	user chooses	yes (after choice)	none (user decides)	high-value edits (docs)

`Resolution`: `Accept(remote)` / `Keep(local)` / `Merged(entity)` / `NeedsUser(local, remote)` — the engine applies the first three; `NeedsUser` routes to UI then re-applies.

33.6 End-to-End Data Flow

Detect + resolve a conflict during pull



Field-level three-way merge



33.7 Complete Kotlin Implementation

Package: `com.streamcart.sync.conflict`.

33.7.1 Versioned entity + resolver contract

```
sealed interface Resolution<out T> {
    data class Accept<T>(val remote: T) : Resolution<T>      // take remote
    data class Keep<T>(val local: T) : Resolution<T>         // keep local
    data class Merged<T>(val value: T) : Resolution<T>       // merged result
    data class NeedsUser<T>(val local: T, val remote: T) : Resolution<T>  // route to UI
}

interface ConflictResolver<T> {
    fun resolve(base: T?, local: T?, remote: T): Resolution<T>
}
```

33.7.2 Last-Writer-Wins (by logical version, ts tiebreaker)

```
class LwwResolver<T : Versioned> : ConflictResolver<T> {
    override fun resolve(base: T?, local: T?, remote: T): Resolution<T> = when {
        local == null -> Resolution.Accept(remote)
        remote.version > local.version -> Resolution.Accept(remote)
        remote.version < local.version -> Resolution.Keep(local)
        else -> if (remote.updatedAt >= local.updatedAt) Resolution.Accept(remote) else
Resolution.Keep(local)  // deterministic tiebreak
    }
}
```

33.7.3 Field-level three-way merge

```
class ProfileMergeResolver : ConflictResolver<Profile> {
    override fun resolve(base: Profile?, local: Profile?, remote: Profile): Resolution<Profile>
{
        if (local == null || base == null) return Resolution.Accept(remote)
        fun <F> pick(sel: (Profile) -> F): F {
            val b = sel(base); val l = sel(local); val r = sel(remote)
            return when {
                l == r -> l                     // agree
                l == b -> r                     // only remote changed -> take remote
                r == b -> l                     // only local changed -> take local
                else -> r                       // both changed same field -> tiebreak
(remote/LWW) or NeedsUser
            }
        }
        val merged = Profile(
            id = remote.id,
            name = pick { it.name },
            bio = pick { it.bio },
            avatarUrl = pick { it.avatarUrl },
            version = maxOf(local.version, remote.version) + 1,
            updatedAt = maxOf(local.updatedAt, remote.updatedAt),
        )
        return Resolution.Merged(merged)
    }
}
```

33.7.4 CRDT examples (convergent by construction)

```
// Watch-progress: max wins (monotonic) -> a CRDT; any order converges.
class MaxProgressResolver : ConflictResolver<WatchProgress> {
    override fun resolve(base: WatchProgress?, local: WatchProgress?, remote: WatchProgress) =
        Resolution.Merged(remote.copy(positionMs = maxOf(local?.positionMs ?: 0,
remote.positionMs)))
}

// Wishlist as an OR-Set: union adds, honor removes by tag -> commutative, convergent.
class OrSetResolver : ConflictResolver<WishlistSet> {
    override fun resolve(base: WishlistSet?, local: WishlistSet?, remote: WishlistSet):
Resolution<WishlistSet> {
        val l = local ?: WishlistSet.EMPTY
        val added = l.addedTags + remote.addedTags            // union of add-tags
        val removed = l.removedTags + remote.removedTags      // union of remove-tags
        val live = added.filterNot { it.id in removed.map(Tag::id) }
        return Resolution.Merged(WishlistSet(live.toSet(), added, removed))
    }
}
```

33.7.5 The sync engine (compose push → pull → detect → resolve → apply)

```
class SyncEngine @Inject constructor(
    private val outbox: OutboxDrainer,                // Ch 31 (push)
    private val delta: DeltaSyncEngine,               // Ch 32 (pull) — delegates apply here
    private val resolvers: ResolverRegistry,          // per-type ConflictResolver
    private val dao: SyncDao, private val db: StreamCartDatabase,
    private val conflictQueue: UserConflictQueue,     // for NeedsUser
) {
    suspend fun sync(): Result<Unit> = runCatching {
        outbox.drain()                                 // push local first (fewer conflicts, Ch
29)
        delta.sync(applyRemote = ::applyRemote)        // pull; apply each change via resolver
    }

    private suspend fun applyRemote(remote: Versioned) = db.withTransaction {
        val local = dao.findById(remote.id)
        val base = dao.findBase(remote.id)             // last-synced snapshot (for 3-way)
        // Fast-forward: no local pending or remote already includes local -> just apply.
        if (local == null || local.syncStatus == SyncStatus.SYNCED) { dao.upsert(remote);
return@withTransaction }
        when (val res = resolvers.forType(remote.type).resolve(base, local, remote)) {
            is Resolution.Accept -> dao.upsert(res.remote)
            is Resolution.Keep -> { /* keep local; local pending will re-push */ }
            is Resolution.Merged -> { dao.upsert(res.value); outbox.enqueueMerge(res.value) }
// push merged up
            is Resolution.NeedsUser -> { dao.setStatus(remote.id, SyncStatus.CONFLICT);
conflictQueue.add(local, remote) }
        }
    }
}
```

33.7.6 Module structure

```
:sync/conflict  ConflictResolver, LwwResolver, FieldMergeResolver, CRDT resolvers,
ResolverRegistry
:sync           SyncEngine (push+pull+resolve), UserConflictQueue
:core:database  Versioned entities (version/clock), base snapshots for 3-way merge
```

33.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Concurrent edits two devices	lost update	detect via version/clock; resolve by policy
Apply order differs per device	divergent final state	convergent strategy (LWW/CRDT/deterministic merge)
Clock skew (timestamp LWW)	wrong winner	logical versions/vector clocks; ts only as tiebreak
Merge then local edits again	re-conflict	push merged (new version); keep local pending if newer
Resolve runs during local write	races SSOT	apply in a transaction; version-gate
Non-commutative "merge"	non-convergence	use CRDTs / three-way deterministic merge
User-resolved pending	stale prompt	snapshot both versions; re-check version on apply

- **Convergence is the correctness invariant:** the resolution function must be **deterministic and order-independent** (LWW by version, CRDT, or deterministic three-way merge) so every device computes the **same** result — otherwise "eventual consistency" is a lie.
- **Vector clocks** distinguish *concurrent* (true conflict) from *causal* (one saw the other → fast-forward) — avoiding false conflicts that plain versions flag.

33.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Concurrent edit (true conflict)	version/clock divergence	resolve by policy (auto/user)	converged, intent preserved
Clock skew	—	logical versions	correct winner
Non-convergent ad-hoc merge	devices differ	switch to CRDT/deterministic 3-way	converge
Lost update (entity LWW)	field loss	field-merge / CRDT	both edits survive
User ignores conflict prompt	stuck CONFLICT	keep local; re-prompt; default policy after timeout	not blocked
Merge loop (A ↔ B re-conflict)	repeated conflicts	version monotonicity; CRDT commutativity	terminates
Delete vs edit conflict	tombstone vs update	policy (e.g., delete-wins / resurrect / user)	defined outcome
Process death mid-resolve	transactional apply	re-run; idempotent (Ch 15)	safe


33.9 Failure Scenarios and Recovery

> **Delete-vs-edit is a conflict too:** decide a policy (delete-wins, resurrect-on- edit, or user-resolved) explicitly — it's a common gap that silently loses either the delete or the edit.

33.10 Scaling the Implementation

- **Level 1.** LWW by version on entity overwrite (simple apps, single-value fields).
- **Level 2.** Per-type resolver registry; server-wins for authoritative data; field-merge for records.
- **Level 3 — production.** Three-way merge with base snapshots; CRDTs for mergeable data (sets/counters/progress); user-resolved for high-value; conflict status UX.
- **Level 4 — large.** Vector clocks for precise causality; delete-vs-edit policy; merged-change re-push; conflict analytics; per-field policies.
- **Level 5 — enterprise.** Full CRDT or **Operational Transform** for real-time collaboration (Ch 63), multi-replica convergence proofs, server-side merge services, conflict audit trails, offline + realtime hybrid.

Avoid overengineering: most apps need LWW or simple field-merge. CRDTs/OT are for genuinely collaborative, concurrently-edited data (docs, shared lists, real-time). Don't build a CRDT framework for a settings screen.

33.11 Alternative Approaches and Trade-offs

Strategy	Convergent	Data loss	Complexity	Use when
LWW (by version)	yes	yes (loser's edit)	low	single low-stakes values; last edit is "right"
Server-wins	yes	yes (local)	low	server is authoritative (prices, inventory)
Client-wins	yes	yes (remote)	low	local device is trusted (rare)
Field-level 3-way merge	yes (deterministic)	minimal	medium	structured records, different fields edited
CRDT / op-merge	yes (by construction)	none	medium–high	collaborative/mergeable (sets, counters, text)
User-resolved	yes (after choice)	none	medium (UX)	high-value edits; auto unsafe
Operational Transform	yes	none	high	real-time collaborative text (Ch 63)

Detection: versions vs vector clocks. A single version counter detects divergence but can't tell *concurrent* from *causal* — it may flag a conflict when one change actually descended from the other (false conflict). **Vector clocks** (a counter per replica) capture the happened-before relation precisely, distinguishing true concurrency from causality, at the cost of storing/transmitting the clock. Versions suffice for 2-party (client+server) LWW; vector clocks shine with many replicas/true multi-device concurrency.

LWW vs CRDT (the core trade-off): LWW is trivial and convergent but **loses** the losing write — fine when only the latest value matters. CRDTs guarantee convergence **and** preserve all operations (no loss) by modeling changes as commutative ops, but require designing the data as a CRDT (counter/set/register) and can grow metadata. For collaborative data where every edit matters, invest in CRDTs; for single values, LWW is enough.

33.12 Security Considerations

- **Server validates and authorizes merges:** a client-proposed merge must be re-validated server-side — a tampered client must not merge in a way that escalates privileges or overwrites others' data (Ch 29).
- **Conflict resolution must not leak data:** showing "remote version" to the user in a user-resolved conflict must respect access (don't reveal another user's content they shouldn't see).
- **Causality integrity:** versions/vector clocks shouldn't be client-forgeable in a way that lets a client suppress legitimate server changes — the backend arbitrates.
- **Encrypt conflicting copies at rest** (Ch 24); clear on logout (Ch 22).
- **Audit trails** for important merges (fintech/collaborative) — who changed what, for dispute resolution.
- Client vs backend: resolution runs on-device for UX, but the **backend is the arbiter** of the authoritative state and must enforce that merges are legal.

33.13 Performance and Resource Optimization

- **Resolution is cheap** (compare versions/fields); CRDT merges are typically O(n) in changed elements — keep metadata bounded (tombstone GC for OR-Sets, etc.).
- **Fast-forward the common case:** most incoming changes have no local conflict — detect and apply directly without invoking the resolver.
- **Store base snapshots** only where three-way merge is used (memory/storage cost).
- **Batch resolution** within a delta page transaction (Ch 32/3).
- **Avoid merge loops** (version monotonicity) that cause repeated re-sync.
- **Profiling:** conflict rate, resolution time, merged-change re-push volume, CRDT metadata size.

33.14 Testing Strategy

Conflict logic is pure and highly testable; the engine is tested with fakes.

```
class LwwResolverTest {
    private val r = LwwResolver<Note>()
    @Test fun `higher version wins`() {
        assertEquals(Resolution.Accept(note(v=2)), r.resolve(note(v=1), note(v=1), note(v=2)))
    }
    @Test fun `deterministic tiebreak on equal version`() {
        val a = r.resolve(null, note(v=1, ts=10), note(v=1, ts=20))
        assertTrue(a is Resolution.Accept)          // same on both devices -> convergent
    }
}

class FieldMergeTest {
    @Test fun `different fields edited both survive`() {
        val base = profile(name="A", bio="x")
        val local = base.copy(name="Ada")           // local changed name
        val remote = base.copy(bio="hi", version=2) // remote changed bio
        val m = (ProfileMergeResolver().resolve(base, local, remote) as Resolution.Merged).value
        assertEquals("Ada", m.name); assertEquals("hi", m.bio)   // no loss
    }
    @Test fun `same field conflict tiebreaks deterministically`() { /* both edit name */ }
}

class CrdtTest {
    @Test fun `max progress converges regardless of order`() {
        val r = MaxProgressResolver()
        val x = r.resolve(null, prog(10), prog(30)); val y = r.resolve(null, prog(30), prog(10))
        assertEquals(30, (x as Resolution.Merged).value.positionMs)
        assertEquals(30, (y as Resolution.Merged).value.positionMs)   // commutative
    }
    @Test fun `OR-Set add and concurrent remove converge`() { /* union adds/removes */ }
}

class SyncEngineConflictTest {
    @Test fun `concurrent edit resolved and merged pushed`() = runTest {
        // seed local pending edit; pull remote edit; assert merged entity + outbox has merge
    }
    @Test fun `convergence: two devices reach same state`() = runTest {
        // apply changes in opposite orders on two in-memory engines; assert equal final state
    }
}
```

What to test: resolver purity (LWW/merge/CRDT), **convergence** (apply in different orders → same state), delete-vs-edit policy, user-resolved routing, engine push→pull→resolve integration, idempotent re-apply.

33.15 Production Observability

- **Metrics:** conflict rate (conflicts / syncs), resolution outcome distribution (LWW/merge/CRDT/user), user-resolution completion rate, data-loss incidents (LWW losers — track to decide if a better strategy is needed), merge-loop detection, divergence incidents (should be 0).
- **Logs:** `conflict{type, strategy, outcome}`, `resolve{entityId_hash, versions}` (no PII content, Ch 28), correlation IDs (Ch 1/84).
- **Debugging:** data loss complaints → LWW on fields that need merge → switch to field-merge/CRDT; devices show different state → non-convergent resolution → deterministic/CRDT; false conflicts → versions flagging causal changes → vector clocks; merge loops → version non-monotonic.
- **Alerting:** divergence or merge-loop detection → correctness incident; conflict- rate spike → concurrent-edit pattern change.
- **Privacy:** conflicting copies are user data — redact; audit important merges.

33.16 System Design Interview Preparation

> 40+ questions.

Fundamentals

1. *What is a sync conflict?* The same entity changed in two replicas (local + remote) since they last agreed.
2. *How detect it?* Versions (base vs server) or vector clocks; local pending + remote advanced → divergence.
3. *Why not wall-clock LWW?* Can't express causality; clock skew picks wrong winner; use logical versions.
4. *What is convergence?* All replicas reach the same final state regardless of order — the eventual-consistency requirement.
5. *Resolution strategies?* LWW, server/client-wins, field-merge, CRDT/op-merge, user-resolved.

Strategies

6. *LWW — pros/cons?* Simple, convergent; **loses** the other edit. Single values.
7. *Server-wins?* Server authoritative; biased local loss (prices/inventory).
8. *Field-level three-way merge?* Compare base/local/remote per field; take non-conflicting changes from both; minimal loss.
9. *When user-resolved?* High-value edits where auto-resolution is unsafe (docs).
10. *CRDT?* Commutative ops → convergence by construction, no loss (counters/sets/ progress/text).

CRDTs / causality

11. *Example CRDTs?* G-Counter (sum), LWW-Register, OR-Set (add/remove), max (progress).
12. *Why do CRDTs converge?* Operations commute — any order yields the same state.
13. *Vector clocks vs versions?* Vector clocks capture causality (concurrent vs causal); versions can't → false conflicts with many replicas.
14. *Watch-progress conflict?* Max-merge (a CRDT) — furthest position wins, convergent.

Merge mechanics

15. *Three-way merge needs?* A base (last-synced) snapshot.
16. *Both edit the same field?* Tiebreak (LWW) or user-resolved.
17. *Delete vs edit?* Explicit policy (delete-wins / resurrect / user) — don't silently drop.

Engine

18. *Compose with push/pull?* Push outbox (Ch 31) → pull delta (Ch 32) → detect vs local pending → resolve → apply (Ch 3).
19. *Why push before pull?* Fewer conflicts; send local first.
20. *Fast-forward?* No local pending / causal remote → apply directly (no resolver).
21. *Merged change?* Push the merged result up (new version).

Convergence/correctness

22. *How guarantee convergence?* Deterministic, order-independent resolution (LWW by version / CRDT / deterministic 3-way).
23. *Merge loops?* Version monotonicity; CRDT commutativity.
24. *Idempotent apply?* Version-gated upsert (Ch 15/32).

Security

25. *Who arbitrates?* Backend validates/authorizes merges; client resolves for UX.
26. *Leak in user-resolved UI?* Respect access to the "remote" version.

Performance

27. *Resolution cost?* Cheap; fast-forward the common case; bound CRDT metadata.
28. *Base snapshots cost?* Only where 3-way merge is used.

Testing

29. *How test convergence?* Apply changes in opposite orders → assert equal state.
30. *How test resolvers?* Pure functions — LWW/merge/CRDT cases.
31. *How test delete-vs-edit?* Assert the chosen policy outcome.

Production debugging

32. *Data loss complaints?* LWW where merge is needed → field-merge/CRDT.
33. *Devices differ?* Non-convergent resolution.
34. *False conflicts?* Versions flag causal changes → vector clocks.
35. *Merge loops?* Non-monotonic versions.

Design

36. *Real-time collaborative text?* CRDT or OT (Ch 63).
37. *When is LWW acceptable?* Single values where latest wins.
38. *When CRDT worth it?* Concurrently-edited mergeable data.
39. *Multi-device counters?* G-Counter (per-replica sums).
40. *Audit important merges?* Trails for fintech/collaborative.

Design-change challenges

1. **Two devices edit different profile fields.** Field-level three-way merge (no loss).
2. **Shared list concurrently edited.** OR-Set CRDT (union adds, honor removes).
3. **Watch-progress across devices.** Max-merge CRDT.
4. **Important doc edits conflict.** User-resolved with both versions.
5. **Real-time collaborative editing.** CRDT/OT over WebSocket (Ch 63).

Mock walkthrough (abridged)

Clarify (data semantics, multi-device, acceptable loss?) → detect via versions/ vector clocks → pick strategy per data (LWW single values, field-merge records, CRDT collaborative, user high-value) → guarantee convergence (deterministic/commutative) → compose push→pull→detect→resolve→apply (engine) → delete-vs-edit policy → backend arbitrates → testing convergence.

33.17 Practical Exercises

1. **Beginner.** Implement version-based LWW resolution with a deterministic tiebreaker; prove both devices pick the same winner. *Eval:* convergent.
2. **Intermediate.** Implement a field-level three-way merge (base/local/remote) so edits to different fields both survive. *Eval:* no loss; same-field tiebreak.
3. **Advanced.** Implement a max-progress and an OR-Set CRDT; prove commutativity (order-independent). *Eval:* same state regardless of order.
4. **Production debugging.** Fix a sync using entity-level LWW that loses concurrent field edits and whose two devices diverge. *Eval:* field-merge/CRDT; converges.
5. **System design challenge.** Build the sync engine (push→pull→detect→resolve→ apply) with a per-type resolver registry and user-resolved routing. *Eval:* auto where safe; prompts for true conflicts; converges.

(Solutions withheld unless requested.)

33.18 Chapter Summary & Reference Sheet

Key decisions: detect conflicts with **logical versions / vector clocks** (not wall clocks); **resolve** with a strategy matched to the data — **LWW** for single low-stakes values, **server-wins** for authoritative data, **field-level three-way merge** for records, **CRDTs** for collaborative/mergeable data (sets, counters, progress), **user-resolved** for high-value edits; **guarantee convergence** (deterministic, order-independent resolution); handle **delete-vs-edit** explicitly; **compose** detect→resolve with **push (Ch 31) → pull (Ch 32)** into one **sync engine**, **fast-forwarding** the non-conflicting common case and **pushing merged results** up; the **backend arbitrates/validates** merges. **Convergence + intent preservation** are the bars.

Concepts: conflict detection, versions vs vector clocks, causality/ happened-before, convergence/eventual consistency, LWW, server/client-wins, three-way merge, CRDTs (G-Counter/LWW-Register/OR-Set/max), operational transform, user-resolved, delete-vs-edit, sync engine composition.

Common mistakes: wall-clock LWW (skew/causality bugs); entity-level overwrite losing concurrent field edits; non-convergent ad-hoc merges (devices diverge); false conflicts from plain versions (use vector clocks); ignoring delete-vs-edit; auto-resolving high-value edits that need the user; non-idempotent/non-version-gated apply; unbounded CRDT metadata.

Checklist:

- [ ] version/vector-clock detection
- [ ] per-type resolution policy
- [ ] convergent (deterministic/commutative)
- [ ] three-way merge where fields differ
- [ ] CRDTs for collaborative data
- [ ] user-resolved for high-value
- [ ] delete-vs-edit policy
- [ ] fast-forward common case
- [ ] push merged up
- [ ] backend validates merges
- [ ] test convergence (order-independence).

Key APIs/patterns: `Versioned` (version/vectorClock/updatedAt), `ConflictResolver`

- `Resolution` (Accept/Keep/Merged/NeedsUser), LWW/field-merge/CRDT resolvers, `ResolverRegistry`, three-way merge, `SyncEngine` (push+pull+resolve+apply), `db.withTransaction`, idempotent version-gated upsert (Ch 15/32).

References:

- CRDTs (intro): https://crdt.tech/
- Vector clocks / happened-before: https://en.wikipedia.org/wiki/Vector_clock
- Offline-first conflict handling: https://developer.android.com/topic/architecture/data-layer/offline-first#conflict-resolution
- Operational Transformation (real-time, Ch 63): https://en.wikipedia.org/wiki/Operational_transformation
- (composes) Ch 29 Offline-First, Ch 31 Outbox, Ch 32 Delta Sync, Ch 15 Idempotency

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; resolver contract + LWW + 3-way merge + CRDTs + sync engine composition; concurrency (convergence invariant, vector clocks); failure (delete-vs-edit, merge loops); 5 levels; versions-vs-vector-clocks and LWW-vs-CRDT trade-offs; security (backend arbitrates); perf (fast-forward, bounded CRDT); convergence tests; observability (conflict/divergence); 40 Q&A + 5 challenges. ✅
- **Expand later:** real-time collaboration/OT (Ch 63), WebSocket push (Ch 63), advanced CRDTs, audit trails. **Completes Part IV (Ch 29–33).**
- **Interview coverage:** exceeds ≥40 bar.